Open Source Audit Project

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Course: Open Source Software
Software Chosen: LibreOffice

Part A — Origin and Philosophy

A1: Origin of LibreOffice

LibreOffice was created when developers lost trust in OpenOffice after Oracle took control. Earlier, OpenOffice was managed by Sun Microsystems. When Oracle bought Sun, the community felt that the project was no longer safe and open. So they decided to create LibreOffice as a new project that is controlled by the community.

A2: License

LibreOffice uses the MPL license.

The four freedoms are:

- Freedom to use the software
- Freedom to study the code
- Freedom to modify it
- Freedom to share it

Companies can use this software but must follow license rules.

GPL is stricter because it forces sharing changes, while MIT is more open.

Free software means freedom, not just free cost.

A3: Ethics

Not all software should be open source. Some companies need profit. But important tools should be open.

Companies using open source is okay, but they should not take full control.

Open source helps people learn and build better software.

Part B — Linux Footprint

I installed LibreOffice using:

sudo apt install libreoffice

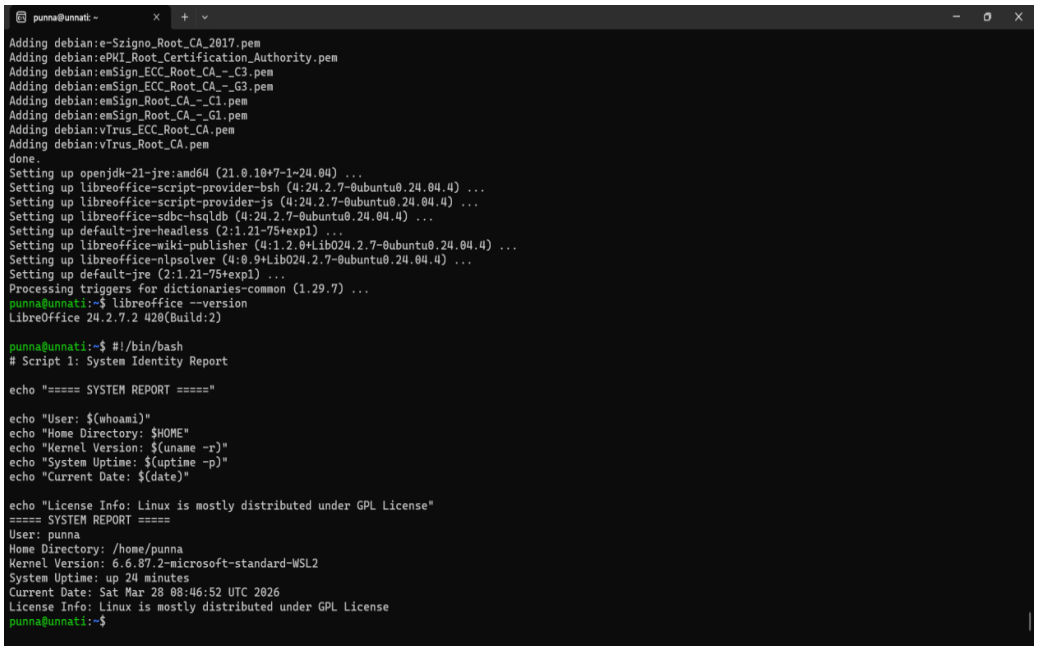
The software is located in:

- /usr/bin/libreoffice
- ~/.config/libreoffice

It runs under the normal user.

Commands used:

- which libreoffice
- whoami



Part C — Ecosystem

LibreOffice depends on libraries like GTK.

It is used in Linux systems and many organizations.

It is managed by The Document Foundation and supported by a global community.

Part D — Comparison

Feature	LibreOffice	MS Office
Cost	Free	Paid
Security	Open code	Closed
Modify	Yes	No
Control	Community	Company

Conclusion: LibreOffice is good for daily use, but MS Office has more advanced features.

Scripts

Script 1:

```
(#!/bin/bash
# Script 1: System Identity Report
echo "===== SYSTEM REPORT ======"
echo "User: $(whoami)"
echo "Home Directory: $HOME"
echo "Kernel Version: $(uname -r)"
echo "System Uptime: $(uptime -p)"
echo "Current Date: $(date)"
```

echo "License Info: Linux is mostly distributed under GPL License")

```
punna@unnati: ~
Adding debian:e-Szigno_Root_CA_2017.pem
Adding debian:ePKI_Root_Certification_Authority.pem
Adding debian:emSign_ECC_Root_CA_-_C3.pem
Adding debian:emSign_ECC_Root_CA_-_G3.pem
Adding debian:emSign_Root_CA_-_C1.pem
Adding debian:emSign_Root_CA_-_G1.pem
Adding debian:vTrus_ECC_Root_CA.pem
Adding debian:vTrus_Root_CA.pem
done.
Setting up openjdk-21-jre:amd64 (21.0.10+7-1~24.04) ...
Setting up libreoffice-script-provider-bsh (4:24.2.7-0ubuntu0.24.04.4) ...
Setting up libreoffice-script-provider-js (4:24.2.7-0ubuntu0.24.04.4) ...
Setting up libreoffice-sdbc-hsqldb (4:24.2.7-0ubuntu0.24.04.4) ...
Setting up default-jre-headless (2:1.21-75+exp1) ...
Setting up libreoffice-wiki-publisher (4:1.2.0+LibO24.2.7-0ubuntu0.24.04.4) ...
Setting up libreoffice-nlpsolver (4:0.9+LibO24.2.7-0ubuntu0.24.04.4) ...
Setting up default-jre (2:1.21-75+exp1) ...
Processing triggers for dictionaries-common (1.29.7) ...
punna@unnati:~$ libreoffice --version
LibreOffice 24.2.7.2 420(Build:2)

punna@unnati:~$ #!/bin/bash
# Script 1: System Identity Report

echo "===== SYSTEM REPORT ====="

echo "User: $(whoami)"
echo "Home Directory: $HOME"
echo "Kernel Version: $(uname -r)"
echo "System Uptime: $(uptime -p)"
echo "Current Date: $(date)"

echo "License Info: Linux is mostly distributed under GPL License"
===== SYSTEM REPORT =====
User: punna
Home Directory: /home/punna
Kernel Version: 6.6.87.2-microsoft-standard-WSL2
System Uptime: up 24 minutes
Current Date: Sat Mar 28 08:46:52 UTC 2026
License Info: Linux is mostly distributed under GPL License
punna@unnati:~$
```

This script shows system details like user, kernel, and uptime.

Script 2:

```
#!/bin/bash

# Script 2: Package Inspector

PACKAGE="libreoffice"

if dpkg -l | grep -q $PACKAGE; then

   echo "$PACKAGE is installed"

   libreoffice --version

else

   echo "$PACKAGE is NOT installed"

fi
```

echo "LibreOffice is a free and open-source office suite."

```
punna@unnati:~$ #!/bin/bash
# Script 2: Package Inspector

PACKAGE="libreoffice"

if dpkg -l | grep -q $PACKAGE; then
    echo "$PACKAGE is installed"
    libreoffice --version
else
    echo "$PACKAGE is NOT installed"
fi

echo "LibreOffice is a free and open-source office suite."
libreoffice is installed
LibreOffice 24.2.7.2 420(Build:2)

LibreOffice is a free and open-source office suite.
punna@unnati:~$
```

This script checks if LibreOffice is installed.

Script 3:

```
#!/bin/bash

echo "===== DIRECTORY CHECK ======"

DIRS=("/etc" "/var/log" "$HOME")

for DIR in "${DIRS[@]}"

do

   echo "Checking: $DIR"

   du -sh $DIR

   ls -ld $DIR

   echo "----------------------"

done

CONFIG="$HOME/.config/libreoffice"

if [ -d "$CONFIG" ]; then

   echo "LibreOffice config folder found:"

   ls -ld $CONFIG
```

else

echo "LibreOffice config folder not found"

fi

```
punna@unnati:~$ ./script3.sh
punna@unnati:~$ #!/bin/bash

echo "===== DIRECTORY CHECK ====="

DIRS=("/etc" "/var/log" "$HOME")

for DIR in "${DIRS[@]}"
do
    echo "Checking: $DIR"
    du -sh $DIR
    ls -ld $DIR
    echo "----------------------"
done

CONFIG="$HOME/.config/libreoffice"

if [ -d "$CONFIG" ]; then
    echo "LibreOffice config folder found:"
    ls -ld $CONFIG
else
    echo "LibreOffice config folder not found"
fi
===== DIRECTORY CHECK =====
Checking: /etc
du: cannot read directory '/etc/ssl/private': Permission denied
du: cannot read directory '/etc/polkit-1/rules.d': Permission denied
du: cannot read directory '/etc/credstore.encrypted': Permission denied
du: cannot read directory '/etc/credstore': Permission denied
9.0M    /etc
drwxr-xr-x 97 root root 4096 Mar 28 14:02 /etc
----------------------
Checking: /var/log
du: cannot read directory '/var/log/private': Permission denied
50M     /var/log
drwxrwxr-x 9 root syslog 4096 Mar 28 14:02 /var/log
----------------------
Checking: /home/punna
660K    /home/punna
drwxr-x--- 4 punna punna 4096 Mar 28 09:25 /home/punna
----------------------
LibreOffice config folder found:
drwxr-xr-x 3 punna punna 4096 Mar 28 14:22 /home/punna/.config/libreoffice
punna@unnati:~$ |
```

This script checks directory size and permissions.

Script 4:

#!/bin/bash

LOGFILE="/var/log/syslog"

COUNT=0

```
if [ ! -f "$LOGFILE" ]; then
    echo "Log file not found"
    exit 1
fi
while read line
do
    if echo "$line" | grep -i "error" > /dev/null
    then
        COUNT=$((COUNT+1))
    fi
done < "$LOGFILE"
echo "Total error lines in log file: $COUNT"
```

```
punna@unnati:~$ #!/bin/bash

LOGFILE="/var/log/syslog"
COUNT=0

if [ ! -f "$LOGFILE" ]; then
    echo "Log file not found"
    exit 1
fi

while read line
do
    if echo "$line" | grep -i "error" > /dev/null
    then
        COUNT=$((COUNT+1))
    fi
done < "$LOGFILE"

echo "Total error lines in log file: $COUNT"
Total error lines in log file: 17
punna@unnati:~$
```

This script counts errors in log files.

Script 5:

#!/bin/bash

read -p "Enter tool you use daily: " TOOL

read -p "What does freedom mean to you? " FREEDOM

read -p "What will you build? " BUILDecho ""

echo "===== YOUR OPEN SOURCE MANIFESTO ====="

echo "I use $TOOL every day."

echo "For me, freedom means $FREEDOM."

echo "I want to build $BUILD and share it with everyone."

```
punna@unnati:~$ #!/bin/bash

read -p "Enter tool you use daily: " TOOL
read -p "What does freedom mean to you? " FREEDOM
read -p "What will you build? " BUILD

echo ""
echo "===== YOUR OPEN SOURCE MANIFESTO ====="
echo "I use $TOOL every day."
echo "For me, freedom means $FREEDOM."
echo "I want to build $BUILD and share it with everyone."
Enter tool you use daily: LibreOffice
What does freedom mean to you? ability to use and modify software freely
What will you build? open source applications for students

===== YOUR OPEN SOURCE MANIFESTO =====
I use LibreOffice every day.
For me, freedom means ability to use and modify software freely.
I want to build open source applications for students and share it with everyone.
punna@unnati:~$
```

This script creates a personal open source statement.

Conclusion

This project helped me understand open source software, how it works, and why it is important. LibreOffice shows how community-driven software can be successful.